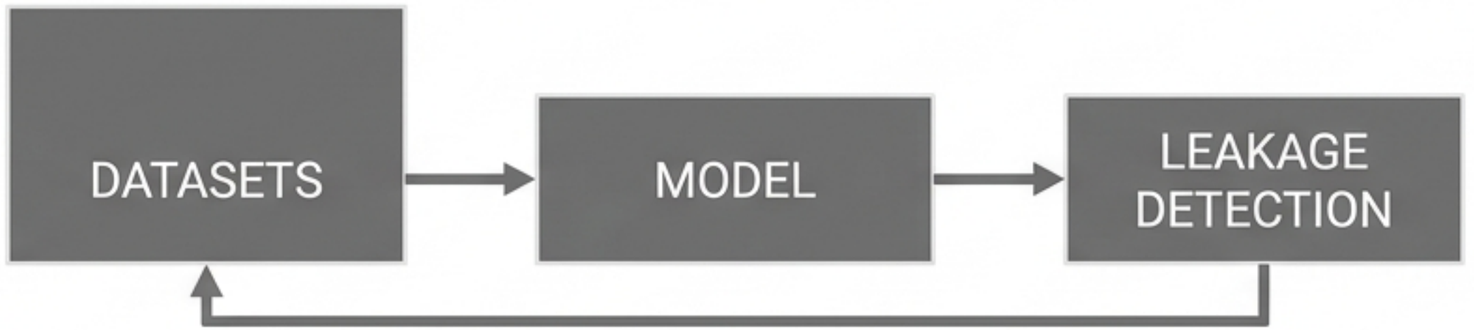
Module 10, Lesson 3
AI 101 for Mechanical Engineers
Leakage is information crossing the wrong boundary

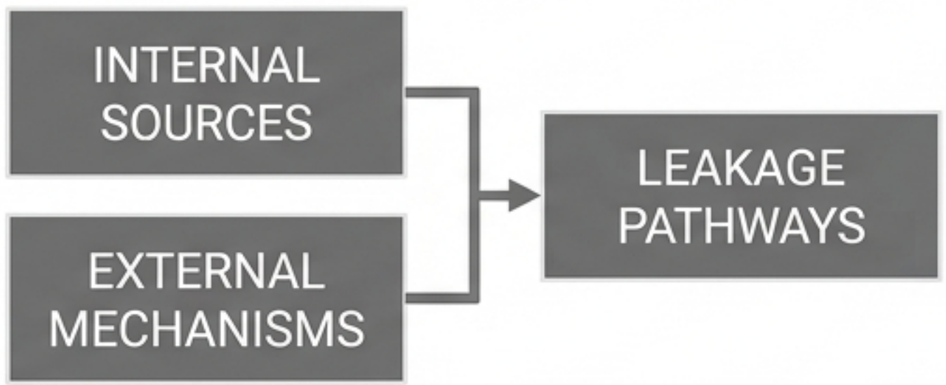
Gemini Notebook

1. Fundamentals of Data Leakage

2. Sources and Mechanisms of Leakage

3. Evaluation and Reproducibility







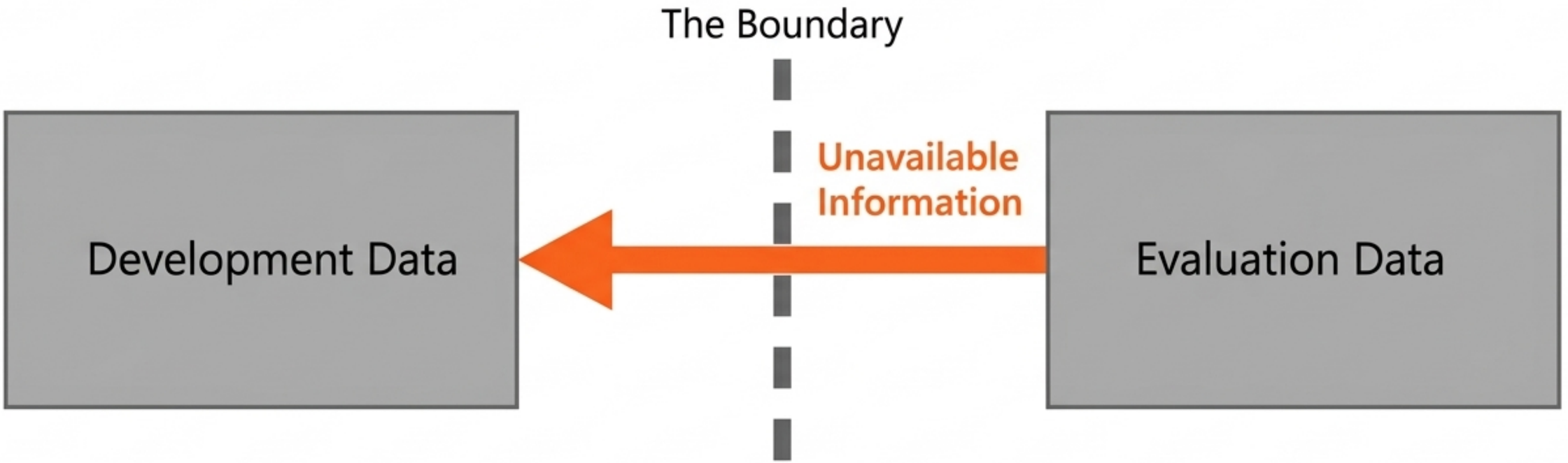
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1 Fundamentals of Data Leakage

Purpose: Define data leakage and establish strict information boundaries in predictive modeling modeling.

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Defining Data Leakage



- Leakage is information crossing an incorrect system boundary.	- Evaluation data is mistakenly accessed during training phases.
- Unavailable information improperly influences model development.	- It invalidates the accuracy of the intended prediction.

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2

Sources and Mechanisms of Leakage

Purpose: Identify specific pathways for target, preprocessing, and future data leakage.

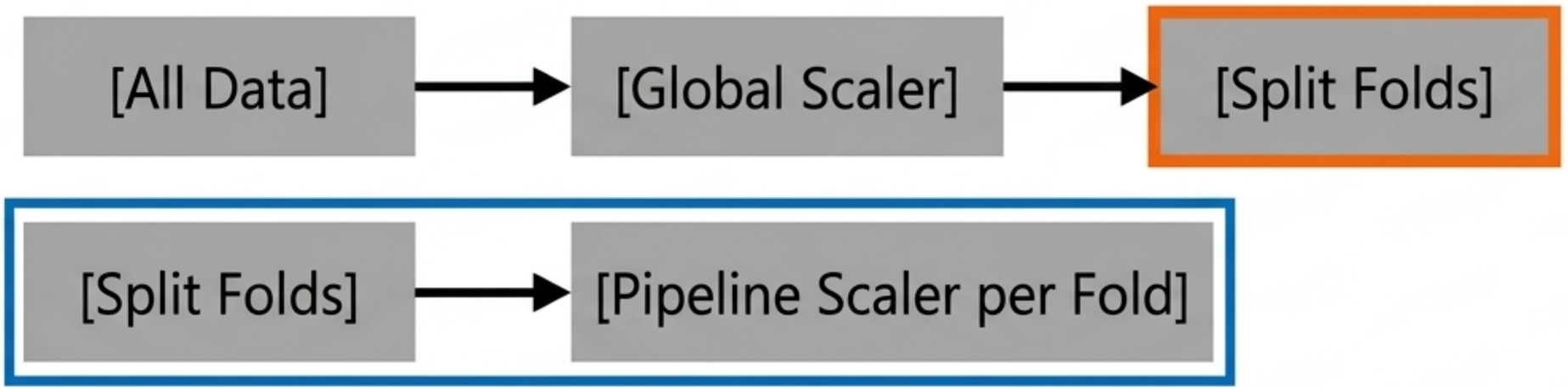
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Taxonomy of Leakage Routes

Leakage Route	Cause of Boundary Crossing	Category
Post-failure repair code	Not available before failure	Target-derived
Global fitted scaler	Held-out data influence fitting	Preprocessing
Duplicate events	Evaluation shares event information	Set duplication
Centered window	Contains future sensor readings	Temporal

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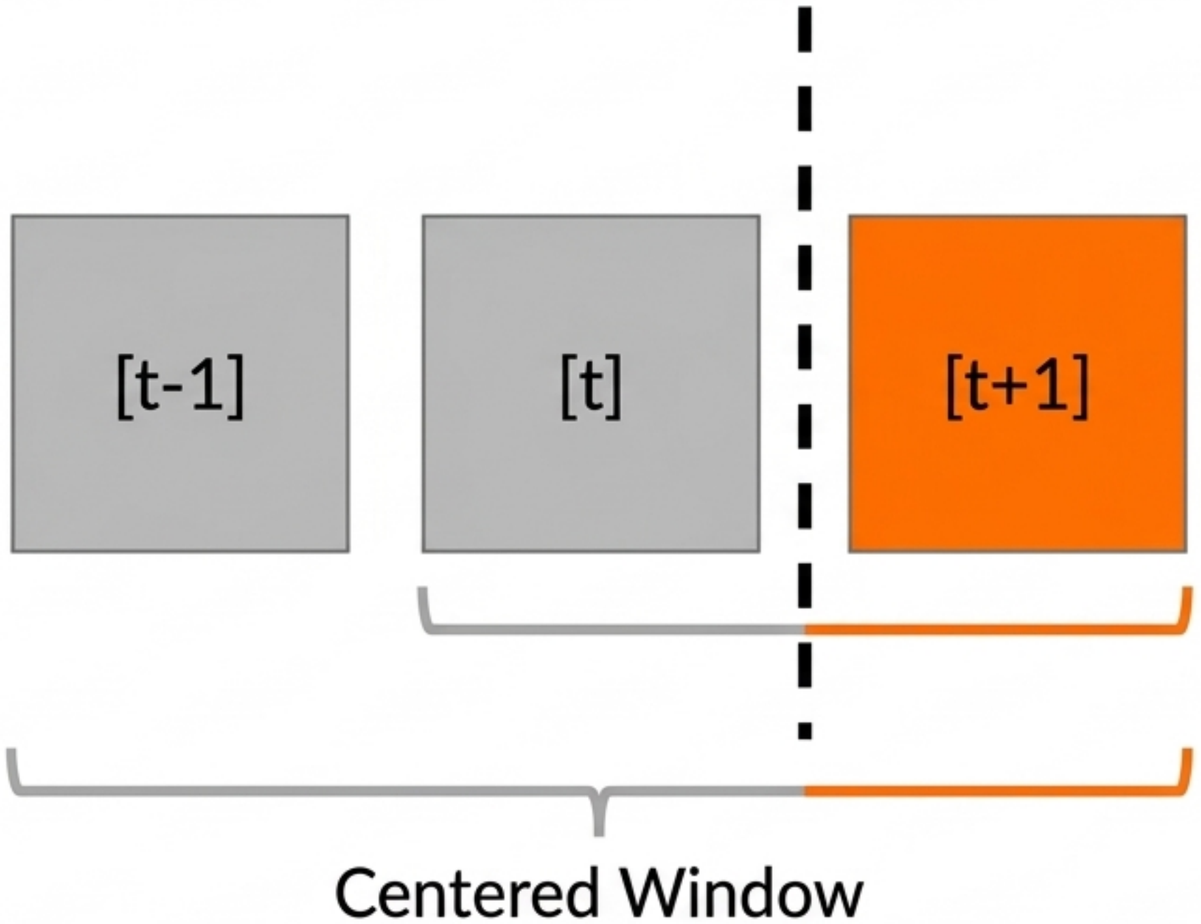
Preprocessing Leakage Mechanism



- Scaling datasets before splitting introduces preprocessing leakage.
- Held-out evaluation folds improperly influence the global scaler.
- The evaluated procedure differs from the intended trained procedure.
- Utilize data pipelines to scale only within isolated folds.
- Training-fold-only imputation respects the information boundary.

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Temporal Leakage: Moving Averages



- A centered moving average feature requires future observations.
- Live alarms strictly operate at the present time t.
- Calculations at t cannot utilize measurements from t+1.
- Chronological dataset splitting alone does not repair temporal feature leakage.

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1. Fundamentals of Data Leakage
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3 Evaluation and Reproducibility

Purpose: Establish protocols for honest evidence recording and reliable evaluation procedures.

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1. Fundamentals of Data Leakage
2. Sources and Mechanisms of Leakage
3. Evaluation and Reproducibility

Honest Evidence Recording

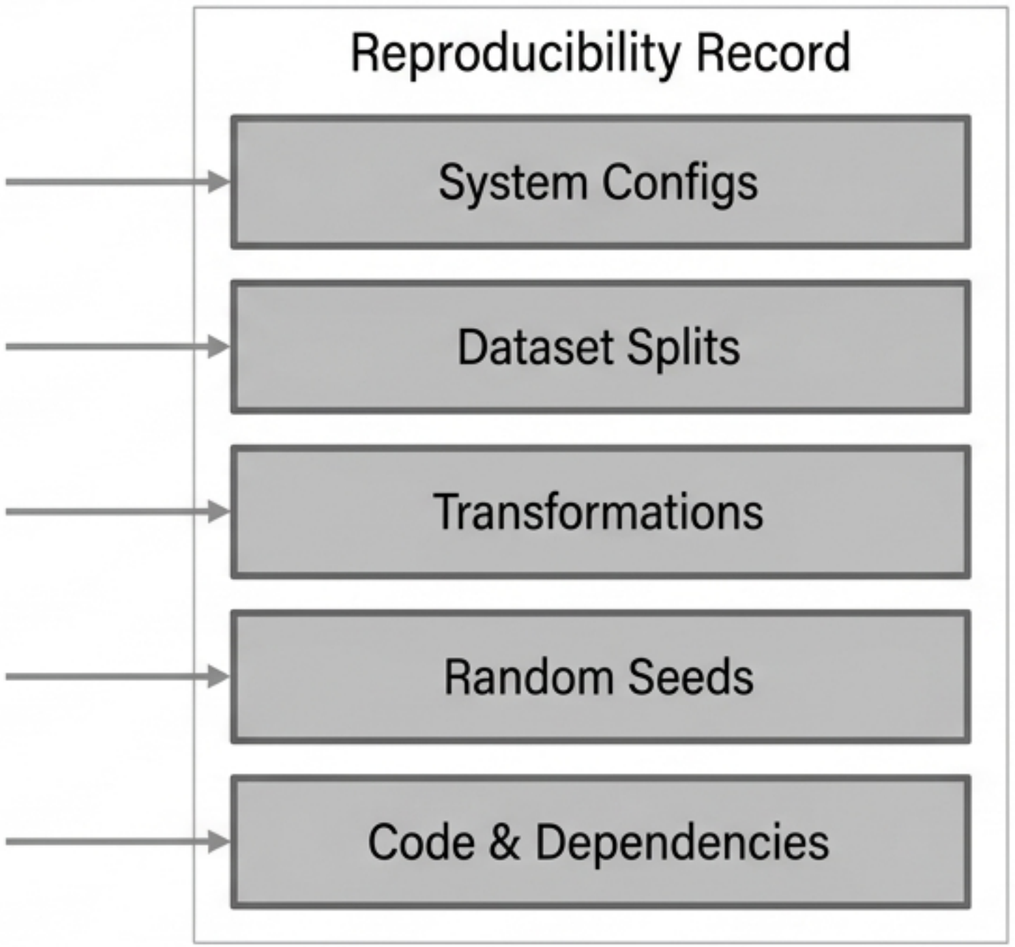


- Final evaluation testing must occur strictly once.
- Engineers must report the exact procedure executed.
- Reusing test data invalidates the independence of the evaluation.
- Limitation disclosures are mandatory if test sets are reused.
- Obtain genuinely new evidence whenever test reuse occurs.

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Reproducibility Requirements

- Reproducibility requires documenting exact system configurations.
- Record all dataset split identifiers.
- Document all mathematical transformations applied to data.
- Log random state seeds used in initialization.
- Archive specific source code and software dependency versions.



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Key Points

1. Data leakage improperly uses unavailable evaluation information during development.
2. Leakage routes include global preprocessing, duplicate events, and temporal windows.
3. Valid evaluation demands single-use testing, procedural transparency, and reproducibility records.

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